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BATTERY AND ELECTRICAL EQUIPMENT

Abstract

Provided are a battery and an electrical equipment. The battery includes a battery pack, a first thermal management component, and thermal insulating member. The battery pack comprises multiple battery cells; along a first direction, the first thermal management component is arranged on one side of the battery pack, and the first thermal management component is configured to manage a temperature of the battery pack. Along the first direction, the thermal insulating member is positioned on the side of the first thermal management component facing away from the battery pack. A thermal insulating member is provided on the first thermal management component, and the thermal insulating member is located on the side of the first thermal management component facing away from the battery pack in the first direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/CN2023/073601, filed on Jan. 8, 2023, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of battery technology, in particular to a battery, and an electrical equipment.

BACKGROUND

[0003] In recent years, new energy vehicles have made significant progress. In the field of electric vehicles, power batteries play an irreplaceable and important role as the power source of electric vehicles. The battery consists of a box and multiple battery cells housed within the box. As a core component of new energy vehicles, batteries have high requirements in terms of use performance and service life. At present, in order to optimize the internal space for the existing batteries, it is usually necessary to stack multiple battery cells and connect busbars and other components between them when assembling them into a box, so as to achieve series or parallel connection between multiple battery cells. However, during use, the battery cells inside the battery generate a large amount of heat during continuous charging and discharging, which can cause a sharp rise in the internal temperature of the battery, seriously affecting its use performance and service life, and even leading to significant safety hazards during its use, which is not conducive to consumer safety.

SUMMARY

[0004] The embodiments of the present disclosure provide a battery and an electrical equipment, which can effectively improve the use performance and service life of the battery.

[0005] In the first aspect, the embodiments of the present disclosure provide a battery, comprising a battery pack, a first thermal management component, and a thermal insulating member, wherein the battery pack comprises multiple battery cells; along a first direction, the first thermal management component is arranged on one side of the battery pack, and the first thermal management component is configured to manage a temperature of the battery pack; and along the first direction, the thermal insulating member is arranged on one side of the first thermal management component facing away from the battery pack.

[0006] In embodiments such as above, a first thermal management component is provided on one side of the battery pack in the first direction to manage the temperature of the battery cells of the battery pack through the first thermal management component. A thermal insulating member is provided on the first thermal management component, and the thermal insulating member is located on the side of the first thermal management component facing away from the battery pack in the first direction, so as to reduce the phenomenon of heat exchange between the first thermal management component and other components or the outside world, thereby reducing the energy loss of the first thermal management component and ensuring the thermal management ability of the first thermal management component on the battery cells. This can effectively regulate the temperature of the battery cells, reduce the temperature rise phenomenon of the battery cells, and improve the use performance and service life of the battery.

[0007] In some embodiments, along the first direction, both sides of the battery pack are provided with the first thermal management component, and each first thermal management component is provided with the thermal insulating member on the side facing away from the battery pack.

[0008] In embodiments such as above, the battery pack is equipped with first thermal management

components on both sides in the first direction, and each first thermal management component is correspondingly equipped with a thermal insulating member, which can further improve the thermal management ability of the first thermal management component on the battery cells of the battery pack, reduce the energy loss of the first thermal management components located on both sides of the battery pack, and further enhance the use performance and service life of the battery. [0009] In some embodiments, the first thermal management component has a first surface facing away from the battery pack along the first direction; and the thermal insulating member is connected to the first surface and covers at least a portion of the first surface.

[0010] In embodiments such as above, by connecting the thermal insulating member to the first surface of the first thermal management component and covering at least a part of the first surface, it is convenient to assemble the thermal insulating member on the one hand, which is conducive to reducing assembly difficulty. On the other hand, it can improve the thermal insulation effect of the thermal insulating member on the first thermal management component, thereby reducing energy loss of the first thermal management component.

[0011] In some embodiments, the thermal insulating member is a thermal insulating pad, and a thermal conductivity coefficient of a material of the thermal insulating member is k , satisfying a condition of $k \leq 0.5 \text{ W/(mK)}$.

[0012] In embodiments such as above, the thermal insulating member is a thermal insulating pad made of a material with a thermal conductivity coefficient less than or equal to 0.5 W/(mK) , which provides good thermal insulation effect on the first thermal management component. The thermal insulating member in this structure is conducive to reducing the assembly difficulty between the thermal insulating member and the first thermal management component, as well as reducing the manufacturing cost of the thermal insulating member.

[0013] In some embodiments, the thermal insulating member has a second surface facing away from the first thermal management component along the first direction, and multiple grooves are provided on the second surface.

[0014] In embodiments such as above, multiple grooves are provided on the second surface of the thermal insulating member facing away from the first thermal management component to reduce the contact area between the second surface and other components, and to separate the thermal insulating member from other components through the grooves, thereby further improving the thermal insulation effect of the thermal insulating member on the first thermal management component.

[0015] In some embodiments, the grooves penetrate both ends of the thermal insulating member along a second direction, wherein the second direction is perpendicular to the first direction.

[0016] In embodiments such as above, by setting the groove as a structure extending along a second direction perpendicular to the first direction, and the groove passing through both ends of the thermal insulating member along the second direction, it is easy to process and manufacture the groove, which is conducive to reducing the difficulty of processing the thermal insulating member.

[0017] In some embodiments, a thickness of the thermal insulating member along the first direction is $D_{\text{sub}.1}$, satisfying a condition of $0.1 \text{ mm} \leq D_{\text{sub}.1} \leq 5 \text{ mm}$.

[0018] In embodiments such as above, by setting the thickness of the thermal insulating member in the first direction to 0.1 mm to 5 mm , on the one hand, it can avoid the phenomenon of poor thermal insulation effect of the thermal insulating member caused by the thickness of the thermal insulating member being too small, and on the other hand, it can avoid the phenomenon of the thermal insulating member occupying too much internal space of the battery due to the thickness of the thermal insulating member being too large, thereby effectively reducing the volume of the battery and improving the energy density of the battery.

[0019] In some embodiments, a thermal insulating chamber is formed inside the thermal insulating member.

[0020] In embodiments such as above, a thermal insulating chamber is arranged inside the thermal

insulating member to achieve thermal insulation effect through the air contained in the thermal insulating chamber. The structure is simple and easy to manufacture.

[0021] In some embodiments, the wall thickness of the thermal insulating chamber along the first direction is $D_{sub.2}$, satisfying a condition of $0.1\text{ mm} \leq D_{sub.2} \leq 3\text{ mm}$.

[0022] In embodiments such as above, by setting the thickness of the wall of the thermal insulating chamber in the first direction to 0.1 mm to 3 mm, on the one hand, it can avoid the phenomenon of weak structural strength of the thermal insulating member caused by the thickness of the wall of the thermal insulating chamber being too small to effectively support the structural form of the thermal insulating chamber, and on the other hand, it can avoid the phenomenon of excessive space occupied by the thermal insulating member caused by the thickness of the wall of the thermal insulating chamber being too large, so as to optimize the internal space of the battery.

[0023] In some embodiments, the thermal insulating member has a reinforcing part, wherein the reinforcing part is located within the thermal insulating chamber; and along the first direction, the reinforcing part is connected to two chamber wall surfaces of the thermal insulating chamber that are facing each other.

[0024] In embodiments such as above, a reinforcing part is also provided inside the thermal insulating chamber of the thermal insulating member, and the reinforcing part is connected to the two chamber wall surfaces of the thermal insulating chamber that are facing each other in the first direction, so as to support the two chamber wall surfaces of the thermal insulating chamber that are facing each other in the first direction through the reinforcing part, thereby enhancing the overall structural strength of the thermal insulating member and ensuring the structural form of the thermal insulating chamber.

[0025] In some embodiments, the thermal insulating member is integrally formed with the first thermal management component.

[0026] In embodiments such as above, by integrally forming the thermal insulating member with the thermal insulating chamber formed inside and the first thermal management component, on the one hand, the stability and reliability of the connection between the thermal insulating member and the first thermal management component can be effectively improved to ensure the thermal insulation effect of the thermal insulating member on the first thermal management component. On the other hand, the assembly process of the thermal insulating member and the first thermal management component can be optimized during the battery assembly process to optimize the production cycle of the battery, which is conducive to improving the production efficiency of the battery.

[0027] In some embodiments, along the first direction, an accommodating groove is provided on one side of the first thermal management component facing away from the battery pack, and the accommodating groove is configured to accommodate at least a portion of the thermal insulating member.

[0028] In embodiments such as above, by setting the accommodating groove on the side of the first thermal management component facing away from the battery pack in the first direction, the thermal insulating member can be accommodated in the accommodating groove. On the one hand, it can play a certain limiting and positioning role for the thermal insulating member, which is conducive to reducing the assembly difficulty between the thermal insulating member and the first thermal management component and improving the production efficiency of the battery. On the other hand, setting the thermal insulating member in the accommodating groove can provide certain protection for the thermal insulating member to avoid the phenomenon of wear or damage to the thermal insulating member during use, thereby improving the service life of the thermal insulating member.

[0029] In some embodiments, multiple battery cells are arranged along the first direction; and the battery pack further comprises a second thermal management component, wherein one second thermal management component is arranged between every two adjacent battery cells along the

first direction, and the second thermal management component is configured to manage a temperature of the battery cells.

[0030] In embodiments such as above, multiple battery cells are arranged along the first direction, and a second thermal management component is also provided between every two adjacent battery cells to manage the temperature of multiple battery cells in the battery pack through the cooperation of the first thermal management component and the second thermal management components, which is conducive to alleviating the temperature rise of the battery during use. In addition, due to the thermal insulating member provided on one side of the first thermal management component facing away from multiple battery cells of the battery pack in the first direction, the phenomenon of heat exchange between the first thermal management component and other components can be reduced. This ensures that the temperature management effects of the first thermal management component and second thermal management component on the battery cells are the same, thereby reducing the temperature difference between multiple battery cells in the battery pack. This is beneficial for ensuring temperature to be even in various areas inside the battery, thereby effectively improving the use performance and service life of the battery.

[0031] In some embodiments, a thickness of the second thermal management component in the first direction is greater than a thickness of the first thermal management component in the first direction.

[0032] In embodiments such as above, by setting the thickness of the second thermal management component to be greater than that of the first thermal management component, the heat transfer capacity of the second thermal management component is greater than that of the first thermal management component. Since the second thermal management component located between two adjacent battery cells in the first direction needs to exchange heat with the battery cells on both sides, while the first thermal management component only needs to exchange heat with the battery cell on one side, batteries with this structure can meet the heat transfer needs of battery cells in different regions, thereby improving the temperature uniformity in various areas inside the battery.

[0033] In some embodiments, the battery comprises multiple battery packs, and each of the battery packs is equipped with the first thermal management component and the thermal insulating member.

[0034] In embodiments such as above, the battery is provided with multiple battery packs, and each of the battery packs is equipped with the first thermal management component and the thermal insulating member, which can effectively ensure the temperature management function inside the battery while improving the energy storage capacity of the battery, thereby facilitating the improvement of the use performance and service life of the battery.

[0035] In some embodiments, the multiple battery packs are arranged at intervals along the first direction.

[0036] In embodiments such as above, by arranging multiple battery packs at intervals along the first direction to facilitate assembly of multiple battery packs, it is beneficial to reduce the assembly difficulty of battery.

[0037] In the second aspect, the embodiment of the present disclosure also provides an electrical equipment, comprising the above battery for providing electrical energy.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0038] In order to more clearly illustrate technical solutions of embodiments of the present disclosure, drawings required for use in the embodiments will be described briefly below. It should be understood that the following drawings are merely illustrative of some embodiments of the present disclosure, which therefore should not be understood as a limitation to the scope. For those

of ordinary skill in the art, other related drawings can also be obtained according to these drawings without any inventive effort.

[0039] FIG. 1 is a schematic view of the structure of the vehicle provided in some embodiments of the present disclosure;

[0040] FIG. 2 is a structure schematic diagram of the battery provided in some embodiments of the present disclosure;

[0041] FIG. 3 is a schematic view of partial structure of the battery provided in some embodiments of the present disclosure;

[0042] FIG. 4 is a partial enlarged view of the A position of the battery shown in FIG. 3;

[0043] FIG. 5 is a connection schematic view of the thermal insulating member and the first thermal management component provided in some embodiments of the present disclosure;

[0044] FIG. 6 is a sectional view of the thermal insulating member connected to the first thermal management component provided in some embodiments of the present disclosure;

[0045] FIG. 7 is a sectional view of the thermal insulating member provided in some embodiments of the present disclosure;

[0046] FIG. 8 is a connection schematic view of the thermal insulating member and the first thermal management component provided in some other embodiments of the present disclosure;

[0047] FIG. 9 is a sectional view of the thermal insulating member connected to the first thermal management component provided in some other embodiments of the present disclosure;

[0048] FIG. 10 is a sectional view of the thermal insulating member provided in some other embodiments of the present disclosure;

[0049] FIG. 11 is a sectional view of the thermal insulating member connected to the first thermal management component provided in some other embodiments of the present disclosure;

[0050] FIG. 12 is a sectional view of the thermal insulating member provided in some other embodiments of the present disclosure;

[0051] FIG. 13 is a sectional view of the thermal insulating member provided in some other embodiments of the present disclosure; and

[0052] FIG. 14 is a sectional view of the thermal insulating member connected to the first thermal management component provided in some other embodiments of the present disclosure.

[0053] Reference signs: **1000**—vehicle; **100**—battery; **10**—battery pack; **11**—battery cell; **12**—second thermal management component; **20**—first thermal management component; **21**—first surface; **22**—accommodating groove; **30**—thermal insulating member; **31**—second surface; **32**—groove; **33**—thermal insulating chamber; **34**—reinforcing part; **40**—box; **41**—first box body; **42**—second box body; **200**—controller; **300**—motor; X—first direction; Y—second direction.

DETAILED DESCRIPTION OF EMBODIMENTS

[0054] In order to make the objectives, technical solutions and advantages of the embodiments of the present disclosure clearer, the technical solutions in the embodiments of the present disclosure will be described clearly below with reference to the drawings in the embodiments of the present disclosure. Obviously, the embodiments described are some of, rather than all of, the embodiments of the present disclosure. All the other embodiments obtained by those of ordinary skill in the art based on the embodiments of the present disclosure without any creative effort shall fall within the scope of protection of the present disclosure.

[0055] Unless otherwise defined, all technical and scientific terms used in the present disclosure have the same meanings as those commonly understood by those skilled in the art to which the present disclosure belongs. The terms used in the description of the present disclosure are merely for the purpose of describing specific embodiments, but are not intended to limit the present disclosure. The terms “comprise/include”, “provide”, and any variations thereof in the description and claims, as well as the brief description of the drawings described above of the present disclosure are intended to cover non-exclusive inclusions. The terms “first”, “second”, etc. in the description and the claims of the present disclosure or the drawings are used to distinguish different

objects, rather than describing specific sequences or primary and secondary relationships.

[0056] In the present disclosure, the phrase “embodiment” mentioned means that the specific features, structures, or characteristics described with reference to the embodiments may be encompassed in at least one embodiment of the present disclosure. The phrase at various locations in the description does not necessarily refer to the same embodiment, or an independent or alternative embodiment exclusive of another embodiment.

[0057] In the description of the present disclosure, it should be noted that unless otherwise explicitly specified and defined, the terms “install”, “link”, “connect” and “attach” should be understood in a broad sense, for example, it may be a fixed connection, a detachable connection, or an integrated connection; and may be a direct connection, an indirect connection by means of an intermediate medium, or internal communication between two elements. For those of ordinary skill in the art, the specific meaning of the foregoing terms in the present disclosure may be understood according to specific circumstances.

[0058] The term “and/or” in the present disclosure is only a description for the association relationship of the associated object, indicating that there can be three types of relationships. For example, A and/or B can indicate three conditions: the presence of only A, the presence of both A and B, and the presence of only B. In addition, the character “/” in the present disclosure generally indicates that the associated objects before and after have an “or” relationship.

[0059] In the embodiments of the present disclosure, the same reference signs represent the same components, and for simplicity, detailed descriptions for the same components are omitted in different embodiments. It should be understood that the thickness, length, width, and other dimensions of various components in the embodiments of the present disclosure shown in the drawings, as well as the overall thickness, length, width, and other dimensions of the integrated device, are only illustrative examples and should not constitute any limitations to the present disclosure.

[0060] The term “multiple (a plurality of)” in the present disclosure refers to two or more (including two).

[0061] In the present disclosure, the battery cell can include a lithium ion secondary battery, a lithium ion primary battery, a lithium sulfur battery, a sodium lithium ion battery, a sodium ion battery or a magnesium ion battery, etc., which is not limited by the embodiments of the present disclosure. The battery cell can be cylindrical, flat, or rectangular, or be of other shapes, which is also not limited by the embodiments of the present disclosure. Battery cells are generally divided into three types according to the packaging method: cylindrical battery cell, square battery cell, and soft pack battery cell, which is also not limited by the embodiments of the present disclosure.

[0062] The battery mentioned in the embodiments of the present disclosure refers to a single physical module that includes one or more battery cells to provide higher voltage and capacity. For example, the batteries mentioned in the present disclosure may include battery modules or battery packs, etc. The battery generally includes a box used to encapsulate one or more battery cells. The box can prevent liquids or other foreign objects from affecting the charging or discharging of battery cells.

[0063] The battery cell includes a shell, electrode assemblies, and electrolyte, and the shell accommodates the electrode assemblies and electrolyte. The electrode assembly is composed of a positive electrode plate, a negative electrode plate, and a separator. Battery cells mainly rely on metal ions moving between the positive and negative electrode plates to operate.

[0064] In recent years, new energy vehicles have made significant progress. In the field of electric vehicles, power batteries play an irreplaceable and important role as the power source of electric vehicles. The battery consists of a box and multiple battery cells housed within the box. The batteries, as the core components of new energy vehicles, have high requirements in terms of use performance and cycle life.

[0065] The inventor found that in general power batteries, in order to obtain sufficient power and

optimize the internal space of the battery, multiple battery cells inside the battery box are usually arranged in a stacking manner to form a battery pack. However, during use, the battery cells of the battery having such structure generate a large amount of heat during the use process of continuous charging and discharging, which can cause a rise in the internal temperature of the battery, and the structure of stacking multiple battery cells will exacerbate the occurrence of this phenomenon, thereby seriously affecting use performance and service life of the battery, and even leading to significant safety hazards during its use, which is not conducive to consumer safety. Therefore, in prior art, multiple cooling plates are usually installed inside the battery box, and arranged at intervals, and the battery cell is placed between two adjacent cooling plates to regulate temperature of the battery cell. However, in the battery having such structure, the outermost cooling plate among multiple cooling plates will exchange heat with the battery box or other components, which will reduce the temperature regulation effect of the cooling plate on the battery cells, resulting in poor thermal management ability of the water-cooled plate on the battery cells. This can easily lead to temperature rise inside the battery, resulting in poor overall use performance of the battery and not conducive to improving the service life of the battery.

[0066] Based on the above considerations, in order to solve the problem of poor use performance and short service life of existing batteries, the inventor has conducted in-depth research and designed a battery, which includes a battery pack, a first thermal management component, and a thermal insulating member. The battery pack comprises multiple battery cells; along a first direction, the first thermal management component is arranged on one side of the battery pack, and the first thermal management component is configured to manage a temperature of the battery pack. Along the first direction, the thermal insulating member is positioned on the side of the first thermal management component facing away from the battery pack.

[0067] In this structure of the battery, a first thermal management component is provided on one side of the battery pack in the first direction to manage the temperature of the battery cells of the battery pack through the first thermal management component. A thermal insulating member is provided on the first thermal management component, and the thermal insulating member is located on the side of the first thermal management component facing away from the battery pack in the first direction, so as to reduce the phenomenon of heat exchange between the first thermal management component and other components or the outside world, thereby reducing the energy loss of the first thermal management component and ensuring the thermal management ability of the first thermal management component on the battery cells. This can effectively regulate the temperature of the battery cells, reduce the temperature rise phenomenon of the battery cells, and improve the use performance and service life of the battery.

[0068] The battery disclosed in the embodiments of the present disclosure can be used in, but is not limited to, electrical equipment such as vehicles, ships, or aircraft. The power system of the electrical equipment can be composed of batteries disclosed in the present disclosure, which is beneficial for reducing the temperature rise phenomenon inside the battery and improving its use performance and service life.

[0069] The embodiment of the present disclosure provides an electrical equipment using a battery as a power source, which can be but is not limited to mobile phones, tablet computers, laptops, electric toys, electric tools, electromobile, electric vehicles, ships, spacecraft, and so on. In the above, electric toys can include fixed or mobile electric toys, such as game consoles, electric car toys, electric ship toys and electric aircraft toys, and spacecraft can include aircraft, rockets, space shuttles and spaceships, etc.

[0070] For the convenience of explanation, the following embodiment takes an electrical device in one embodiment of the present disclosure being the vehicle **1000** as an example to explain.

[0071] Referring to FIG. 1, FIG. 1 is a structural schematic view of the vehicle **1000** provided in some embodiments of the present disclosure. The vehicle **1000** can be a fuel powered vehicle, a gas powered vehicle, or a new energy vehicle. The new energy vehicle can be a pure electric vehicle, a

hybrid electric vehicle, or an extended range vehicle, etc. The interior of vehicle **1000** is equipped with battery **100**, which can be installed at the bottom, head, or tail of vehicle **1000**. Battery **100** can be used for the power supply for the vehicle **1000**. For example, battery **100** can serve as the operating power supply for vehicle **1000**. The vehicle **1000** can also include a controller **200** and a motor **300**, wherein the controller **200** is used to control the battery **100** to supply power to the motor **300**, for example, used for the working power demand during startup, navigation, and running of the vehicle **1000**.

[0072] In some embodiments of the present disclosure, the battery **100** can not only serve as the operating power source for vehicle **1000**, but also as the driving power source for vehicle **1000**, thereby replacing or partially replacing fuel or natural gas to provide driving power for vehicle **1000**.

[0073] According to some embodiments of the present disclosure, referring to FIGS. 2 to 5, FIG. 2 is a structure schematic diagram of the battery **100** provided in some embodiments of the present disclosure; FIG. 3 is a schematic view of partial structure of the battery **100** provided in some embodiments of the present disclosure; FIG. 4 is a partial enlarged view of the A position of the battery **100** shown in FIG. 3; and FIG. 5 is a connection schematic view of the thermal insulating member **30** and the first thermal management component **20** provided in some embodiments of the present disclosure. The present disclosure provides a battery **100**, which includes a battery pack **10**, a first thermal management component **20**, and a thermal insulating member **30**. The battery pack **10** includes multiple battery cells **11**. Along a first direction X, the first thermal management component **20** is arranged on one side of the battery pack **10**, and the first thermal management component **20** is configured to manage a temperature of the battery pack **10**. Along the first direction X, the thermal insulating member **30** is positioned on the side of the first thermal management component **20** facing away from the battery pack **10**.

[0074] In an embodiment, battery **100** may also include a box **40**, which is used to provide assembly space for components such as battery pack **10**, thermal management components, and thermal insulating member **30**. That is, battery pack **10**, thermal management components, and thermal insulating member **30** are all accommodated within box **40**.

[0075] The structure of the box **40** can be various. In some embodiments, as shown in FIGS. 2 and 3, the box **40** may include a first box body **41** and a second box body **42**. The first box body **41** and the second box body **42** are mutually covered, and the first box body **41** and the second box body **42** jointly define an assembly space for accommodating the battery pack **10**. The second box body **42** can be in a hollow structure with an opening at one end, the first box body **41** can be in a plate-shaped structure, and the first box body **41** is covered on the opening side of the second box body **42** to jointly define the assembly space by the first box body **41** and the second box body **42**. Of course, in other embodiments, the first box body **41** and the second box body **42** can also both be in hollow structure with an opening on one side, and the opening side of the first box body **41** is covered on the opening side of the second box body **42**. The box **40** formed by the first box body **41** and the second box body **42** can be of various shapes, such as cylinder, cuboid, etc. As an example, in FIG. 2, the box **40** is in a cuboid structure.

[0076] In battery **100**, one or multiple battery packs **10** can be provided in the box **40**. As an example, in FIG. 3, the battery **100** includes three battery packs **10**, which are arranged at intervals along the first direction X inside the box **40**. Of course, in other embodiments, the battery packs **10** set inside the box **40** can also be provided in the number of two, four, or five, etc.

[0077] Battery pack **10** includes multiple battery cells **11**, and each battery cell **11** can be a secondary battery or a primary battery; and can also be a lithium sulfur battery, sodium ion battery, or magnesium ion battery, but is not limited thereto. The battery cell **11** can be cylindrical, flat, or rectangular, or be of other shapes. As an example, in FIG. 3, the battery cell **11** is in a shape of cuboid.

[0078] Multiple battery cells **11** can be connected in series, parallel, or hybrid. Hybrid connection

refers to both series and parallel connections existing among multiple battery cells **11**. Multiple battery cells **11** can be directly connected in series, parallel, or hybrid together, and then the battery pack **10** composed of multiple battery cells **11** is accommodated in the box **40**. The battery pack **10** may also include other structures, for example, the battery pack **10** may also include a busbar component for achieving electrical connection between multiple battery cells **11**.

[0079] The first thermal management component **20** plays a role in exchanging heat with the battery cell **11** to manage the temperature of the battery cell **11**. For example, the first thermal management component **20** can be a harmonica-shaped tube or the like, and temperature regulation for the battery cell **11** can be achieved through the heat exchange medium contained within the first thermal management component **20**. The specific structure of the first thermal management component **20** can be found in relevant technologies and will not be repeated here.

[0080] Along the first direction X, the first thermal management component **20** is positioned on one side of the battery pack **10**, that is, the first thermal management component **20** is located on one side of the battery pack **10** in the first direction X. It should be noted that the first thermal management component **20** is set correspondingly to the battery pack **10**. If there are multiple battery packs **10**, each battery pack **10** is equipped with the first thermal management component **20**, and the first thermal management component **20** is set on one side of the corresponding battery pack **10** in the first direction X.

[0081] Along the first direction X, the thermal insulating member **30** is positioned on the side of the first thermal management component **20** facing away from the battery pack **10**, that is, the thermal insulating member **30** is positioned on the side of the first thermal management component **20** facing away from the battery pack **10** in the first direction X. That is to say, in the first direction X, the first thermal management component **20** is located between the battery pack **10** and the thermal insulating member **30**. It should be noted that the thermal insulating member **30** is correspondingly arranged with the first thermal management component **20**. If there are multiple battery packs **10**, each thermal insulating member **30** is arranged on the side of the corresponding first thermal management component **20** that faces away from the corresponding battery pack **10** in the first direction X.

[0082] The thermal insulating member **30** is used to separate the first thermal management component **20** from the box **40** or other components of the battery **100**, so as to provide thermal insulation and reduce heat exchange between the first thermal management component **20** and the box **40** or other components of the battery **100**. Optionally, the structure of thermal insulating member **30** can be various. For example, thermal insulating member **30** can be a thermal insulating pad made of thermal insulation material, or a hollow structure with a cavity formed inside to provide thermal insulation through the cavity. When thermal insulating member **30** is a hollow structure, it can be made of metal or non-metal thermal insulation material.

[0083] In some embodiments, multiple battery cells **11** of the battery pack **10** are arranged within the box **40**. For example, in FIGS. 3 and 4, the battery pack **10** includes multiple rows of battery cells **11** arranged along the thickness direction of the battery cells **11**, and each row of battery cells **11** includes multiple battery cells **11** arranged along the length direction of the battery cells **11**.

[0084] In FIGS. 3 and 4, the first direction X is the thickness direction of the battery cells **11**, that is, the battery pack **10** includes multiple rows of battery cells **11** arranged along the first direction X. That is to say, the thickness direction of the first thermal management component **20** is the same as that of the battery cells **11**. It should be noted that in some embodiments, the first direction X can also be the length direction of the battery cell **11** or the height direction of the battery cell **11**, that is, the first thermal management component **20** can also be set on one side of multiple battery cells **11** in the length direction of the battery cell **11**, that is to say, the thickness direction of the first thermal management component **20** is the same as the length direction of the battery cell **11**, or it can be set on one side of multiple battery cells **11** in the height direction of the battery cell **11**, that is to say, the thickness direction of the first thermal management component **20** is perpendicular to

the thickness direction of the battery cell **11**, and the first thermal management component **20** covers the top or bottom of multiple battery cells **11**.

[0085] For example, the battery pack **10** includes multiple rows of battery cells **11** arranged along the first direction X, and each row of battery cells **11** includes multiple battery cells **11** arranged along the second direction Y. The first thermal management component **20** is disposed on one side of the multiple battery cells **11** in the first direction X, and the first direction X and the second direction Y are perpendicular to each other. It should be noted that in other embodiments, the battery pack **10** may also only include multiple battery cells **11** arranged along the first direction X, that is, the battery pack **10** only includes one column of multiple battery cells **11** arranged along the first direction X.

[0086] The first thermal management component **20** is provided on one side of the battery pack **10** in the first direction X to manage the temperature of the battery cells **11** of the battery pack **10** through the first thermal management component **20**. A thermal insulating member **30** is provided on the first thermal management component **20**, and the thermal insulating member **30** is located on the side of the first thermal management component **20** facing away from the battery pack **10** in the first direction X, so as to reduce the phenomenon of heat exchange between the first thermal management component **20** and other components or the outside world, thereby reducing the energy loss of the first thermal management component **20** and ensuring the thermal management ability of the first thermal management component **20** on the battery cells **11**. This can effectively regulate the temperature of the battery cells **11**, reduce the temperature rise phenomenon of the battery cells **11**, and improve the use performance and service life of the battery **100**.

[0087] According to some embodiments of the present disclosure, as shown in FIG. 3, along the first direction X, both sides of the battery pack **10** are provided with the first thermal management component **20**, and each first thermal management component **20** is provided with the thermal insulating member **30** on the side facing away from the battery pack **10**.

[0088] Along the first direction X, both sides of the battery pack **10** are equipped with first thermal management components **20**, that is, each battery pack **10** is correspondingly equipped with first thermal management components **20** on both sides of the first direction X, so that the battery pack **10** is located between the corresponding two first thermal management components **20** in the first direction X.

[0089] Each first thermal management component **20** is provided with a thermal insulating member **30** on the side facing away from the battery pack **10**, that is, each first thermal management component **20** is correspondingly provided with one thermal insulating member **30**, and the thermal insulating member **30** is provided on the side of the corresponding first thermal management component **20** facing away from the corresponding battery pack **10** in the first direction X.

[0090] The battery pack **10** is equipped with first thermal management components **20** on both sides in the first direction X, and each first thermal management component **20** is correspondingly equipped with a thermal insulating member **30**, which can further improve the thermal management ability of the first thermal management component **20** on the battery cells **11** of the battery pack **10**, reduce the energy loss of the first thermal management components **20** located on both sides of the battery pack **10**, and further enhance the use performance and service life of the battery **100**.

[0091] According to some embodiments of the present disclosure, referring to FIG. 5, and further referring to FIG. 6, FIG. 6 is a sectional view of the thermal insulating member **30** connected to the first thermal management component **20** provided in some embodiments of the present disclosure. The first thermal management component **20** has a first surface **21** facing away from the battery pack **10** along the first direction X; and the thermal insulating member **30** is connected to the first surface **21** and covers at least a portion of the first surface **21**.

[0092] The thermal insulating member **30** is connected to the first surface **21** and covers at least a part of the first surface **21**, that is, the thermal insulating member **30** can cover part of the first surface **21**, or can completely cover the first surface **21**.

[0093] For example, the thermal insulating member **30** can be connected to the first surface **21** in various ways, such as bonding, snapping, etc.

[0094] By connecting the thermal insulating member **30** to the first surface **21** of the first thermal management component **20** and covering at least a part of the first surface **21**, it is convenient to assemble the thermal insulating member **30** on the one hand, which is conducive to reducing assembly difficulty. On the other hand, it can improve the thermal insulation effect of the thermal insulating member **30** on the first thermal management component **20**, thereby reducing energy loss of the first thermal management component **20**.

[0095] According to some embodiments of the present disclosure, referring to FIGS. **5** and **6**, and further referring to FIG. **7**, FIG. **7** is a sectional view of the thermal insulating member **30** provided in some embodiments of the present disclosure. The thermal insulating member **30** is a thermal insulating pad, and the thermal conductivity coefficient of the material of thermal insulating member **30** is k , which satisfies $k \leq 0.5 \text{ W/(mK)}$.

[0096] The thermal insulating member **30** is a gasket made of material with heat resistance and insulation effects. By placing the thermal insulating member **30** on the side of the first thermal management component **20** facing away from the battery pack **10**, the thermal insulation effect of the thermal insulating member **30** on the first thermal management component **20** can be achieved.

[0097] For example, the material of the thermal insulating member **30** can be aerogel, rubber, polypropylene or polyamide.

[0098] In the embodiment where the thermal insulating member **30** is a thermal insulating pad, the thermal insulating member **30** can be connected to the first thermal management component **20** by bonding or snapping.

[0099] The thermal insulating member **30** is a thermal insulating pad made of a material with a thermal conductivity coefficient less than or equal to 0.5 W/(mK) , which provides good thermal insulation effect on the first thermal management component **20**. The thermal insulating member **30** in this structure is conducive to reducing the assembly difficulty between the thermal insulating member **30** and the first thermal management component **20**, as well as reducing the manufacturing cost of the thermal insulating member **30**.

[0100] According to some embodiments of the present disclosure, referring to FIGS. **8**, **9** and **10**, FIG. **8** is a connection schematic view of the thermal insulating member **30** and the first thermal management component **20** provided in some other embodiments of the present disclosure; FIG. **9** is a sectional view of the thermal insulating member **30** connected to the first thermal management component **20** provided in some other embodiments of the present disclosure; and FIG. **10** is a sectional view of the thermal insulating member **30** provided in some other embodiments of the present disclosure. The thermal insulating member **30** has a second surface facing **31** away from the first thermal management component **20** along the first direction X , and multiple grooves **32** are provided on the second surface **31**.

[0101] The thermal insulating member **30** has a second surface **31** facing away from the first thermal management component **20**, that is, in the first direction X , the surface of the thermal insulating member **30** farthest from the first thermal management component **20** is the second surface **31**.

[0102] Multiple grooves **32** are provided on the second surface **31**, that is, multiple grooves **32** are formed on the second surface **31** that are recessed in the direction closer to the first thermal management component **20** in the first direction X .

[0103] In an embodiment, the shape of the groove **32** provided on the second surface **31** can be various, such as strip-shaped, circular, or elliptical. For example, in FIGS. **8** and **9**, the groove **32** provided on the second surface **31** is of a strip-shaped structure.

[0104] Multiple grooves **32** are provided on the second surface **31** of the thermal insulating member **30** facing away from the first thermal management component **20** to reduce the contact area between the second surface **31** and other components, and to separate the thermal insulating

member **30** from other components through the grooves **32**, thereby further improving the thermal insulation effect of the thermal insulating member **30** on the first thermal management component **20**.

[0105] In some embodiments, referring to FIGS. **8**, **9** and **10** continuously, the grooves **32** penetrate both ends of the thermal insulating member **30** along a second direction Y, wherein the second direction Y is perpendicular to the first direction X.

[0106] The groove **32** is a strip-shaped structure extending along the second direction Y, and passes through both ends of thermal insulating member **30** along the second direction Y.

[0107] The second direction Y is perpendicular to the first direction X. The second direction Y can be the length direction of the first thermal management component **20** or the width direction of the first thermal management component **20**. For example, in FIG. **8**, the second direction Y is the length direction of the first thermal management component **20**, that is, the groove **32** passes through both ends of the thermal insulating member **30** along the length direction of the first thermal management component **20**, and multiple grooves **32** are arranged at intervals along the width direction of the first thermal management component **20**.

[0108] By setting the groove **32** as a structure extending along a second direction Y perpendicular to the first direction X, and the groove **32** passing through both ends of the thermal insulating member **30** along the second direction Y, it is easy to process and manufacture the groove **32**, which is conducive to reducing the difficulty of processing the thermal insulating member **30**.

[0109] According to some embodiments of the present disclosure, as shown in FIGS. **7** and **10**, a thickness of the thermal insulating member **30** along the first direction X is $D_{sub.1}$, satisfying a condition of $0.1\text{ mm} \leq D_{sub.1} \leq 5\text{ mm}$.

[0110] It should be noted that in the embodiment where grooves **32** are provided on the second surface **31** of the thermal insulating member **30**, the thickness $D_{sub.1}$ of the thermal insulating member **30** is the distance between the second surface **31** of the thermal insulating member **30** with grooves **32** and the surface of the thermal insulating member **30** connected to the first thermal management component **20** in the first direction X.

[0111] By setting the thickness of the thermal insulating member **30** in the first direction X to 0.1 mm to 5 mm, on the one hand, it can avoid the phenomenon of poor thermal insulation effect of the thermal insulating member **30** caused by the thickness of the thermal insulating member **30** being too small, and on the other hand, it can avoid the phenomenon of the thermal insulating member **30** occupying too much internal space of the battery **100** due to the thickness of the thermal insulating member **30** being too large, thereby effectively reducing the volume of the battery **100** and improving the energy density of the battery **100**.

[0112] According to some embodiments of the present disclosure, referring to FIGS. **11** and **12**, FIG. **11** is a sectional view of the thermal insulating member **30** connected to the first thermal management component **20** provided in some other embodiments of the present disclosure; and FIG. **12** is a sectional view of the thermal insulating member **30** provided in some other embodiments of the present disclosure. A thermal insulating chamber **33** is formed inside the thermal insulating member **30**.

[0113] The thermal insulating chamber **33** is formed inside the thermal insulating member **30**, that is, the thermal insulating member **30** is in a hollow structure for thermal insulation through the thermal insulating chamber **33** inside the thermal insulating member **30**. The thermal insulating member **30** achieves thermal insulation by air to achieve thermal insulation effect on the first thermal management component **20**.

[0114] In the embodiment where the thermal insulating chamber **33** is formed inside the thermal insulating member **30**, the material of the thermal insulating member **30** can be metal or non-metal, for example, the material of the thermal insulating member **30** can be metal materials such as aluminum and copper, or non-metal materials such as aerogel and rubber.

[0115] Of course, in this embodiment, the connection method between the thermal insulating

member **30** and the first thermal management component **20** can be various. For example, the thermal insulating member **30** can be connected to the first thermal management component **20** through bonding or snapping, or can be formed as an integral structure with the first thermal management component **20**, such as through stamping or extrusion molding processes. For example, in FIG. **11**, the thermal insulating member **30** and the first thermal management component **20** are in a split structure, and the thermal insulating member **30** is bonded to the first thermal management component **20**.

[0116] The thermal insulating chamber **33** is arranged inside the thermal insulating member **30** to achieve thermal insulation effect through the air contained in the thermal insulating chamber **33**. The structure is simple and easy to manufacture.

[0117] In some embodiments, as shown in FIG. **12**, the wall thickness of the thermal insulating chamber **33** along the first direction X is $D_{sub.2}$, satisfying a condition of $0.1\text{ mm} \leq D_{sub.2} \leq 3\text{ mm}$.

[0118] By setting the thickness of the wall of the thermal insulating chamber **33** in the first direction X to 0.1 mm to 3 mm, on the one hand, it can avoid the phenomenon of weak structural strength of the thermal insulating member **30** caused by the thickness of the wall of the thermal insulating chamber **33** being too small to effectively support the structural form of the thermal insulating chamber **33**, and on the other hand, it can avoid the phenomenon of excessive space occupied by the thermal insulating member **30** caused by the thickness of the wall of the thermal insulating chamber **33** being too large, so as to optimize the internal space of the battery **100**.

[0119] According to some embodiments of the present disclosure, referring to FIG. **13**, FIG. **13** is a sectional view of the thermal insulating member **30** provided in some other embodiments of the present disclosure. The thermal insulating member **30** has a reinforcing part **34**, wherein the reinforcing part **34** is located within the thermal insulating chamber **33**; and along the first direction X, the reinforcing part **34** is connected to two chamber wall surfaces of the thermal insulating chamber **33** that are facing each other.

[0120] The thermal insulating member **30** has a reinforcing part **34**, which is set inside the thermal insulating chamber **33**. That is to say, in the embodiment where the thermal insulating chamber **33** is set inside the thermal insulating member **30**, a reinforcing part **34** for supporting the chamber wall of the thermal insulating chamber **33** can also be set inside the thermal insulating chamber **33** of the thermal insulating member **30**.

[0121] Along the first direction X, the reinforcing part **34** is connected to two chamber wall surfaces of the thermal insulating chamber **33** that are facing each other, that is, the reinforcing part **34** is set between two chamber wall surfaces of the thermal insulating chamber **33** that are facing each other in the first direction X, and the two ends of the reinforcing part **34** are respectively connected to the two chamber wall surfaces of the thermal insulating chamber **33** that are facing each other.

[0122] In an embodiment, the reinforcing part **34** provided in the thermal insulating chamber **33** can be one or multiple. For example, in FIG. **13**, multiple reinforcing parts **34** are provided in the thermal insulating chamber **33**, and the multiple reinforcing parts **34** are spaced apart in the thermal insulating chamber **33**.

[0123] The reinforcing part **34** is also provided inside the thermal insulating chamber **33** of the thermal insulating member **30**, and the reinforcing part **34** is connected to the two chamber wall surfaces of the thermal insulating chamber **33** that are facing each other in the first direction X, so as to support the two chamber wall surfaces of the thermal insulating chamber **33** that are facing each other in the first direction X through the reinforcing part **34**, thereby enhancing the overall structural strength of the thermal insulating member **30** and ensuring the structural form of the thermal insulating chamber **33**.

[0124] In some embodiments, the thermal insulating member **30** is integrally formed with the first thermal management component **20**. That is to say, in the embodiment where the thermal insulating

chamber **33** is formed inside the thermal insulating member **30**, the thermal insulating member **30** may be an integrated structure with the first thermal management component **20**, for example, made by stamping or extrusion molding processes, that is, the material of the thermal insulating member **30** is the same as that of the first thermal management component **20**, and the thermal insulating member **30** plays a thermal insulation role through the thermal insulating chamber **33** formed inside to insulate the first thermal management component **20**.

[0125] By integrally forming the thermal insulating member **30** with the thermal insulating chamber **33** formed inside and the first thermal management component **20**, on the one hand, the stability and reliability of the connection between the thermal insulating member **30** and the first thermal management component **20** can be effectively improved to ensure the thermal insulation effect of the thermal insulating member **30** on the first thermal management component **20**. On the other hand, the assembly process of the thermal insulating member **30** and the first thermal management component **20** can be optimized during the assembly process of the battery **100** to optimize the production cycle of the battery **100**, which is conducive to improving the production efficiency of the battery **100**.

[0126] In accordance with some embodiments of the present disclosure, referring to FIG. **14**, FIG. **14** is a sectional view of the thermal insulating member **30** connected to the first thermal management component **20** provided in some other embodiments of the present disclosure. Along the first direction X, an accommodating groove **22** is provided on one side of the first thermal management component **20** facing away from the battery pack **10**, and the accommodating groove **22** is configured to accommodate at least a portion of the thermal insulating member **30**.

[0127] The accommodating groove **22** is used to accommodate at least a part of the thermal insulating member **30**, that is, the thermal insulating member **30** can be fully accommodated in the accommodating groove **22** or partially accommodated in the accommodating groove **22**. That is to say, in the first direction X, the thermal insulating member **30** can extend beyond the accommodating groove **22** or not exceed the accommodating groove **22**.

[0128] It should be noted that in the embodiment where the first thermal management component **20** has a first surface **21** and the thermal insulating member **30** covers the first surface **21**, if the accommodating groove **22** is provided on the side of the first thermal management component **20** facing away from the battery pack **10**, then the first surface **21** is the bottom surface of the accommodating groove **22**, that is, the thermal insulating member **30** is connected to the bottom surface of the accommodating groove **22**.

[0129] By setting the accommodating groove **22** on the side of the first thermal management component **20** facing away from the battery pack **10** in the first direction X, the thermal insulating member **30** can be accommodated in the accommodating groove **22**. On the one hand, it can play a certain limiting and positioning role for the thermal insulating member **30**, which is conducive to reducing the assembly difficulty between the thermal insulating member **30** and the first thermal management component **20** and improving the production efficiency of the battery **100**. On the other hand, setting the thermal insulating member **30** in the accommodating groove **22** can provide certain protection for the thermal insulating member **30** to avoid the phenomenon of wear or damage to the thermal insulating member **30** during use, thereby improving the service life of the thermal insulating member **30**.

[0130] According to some embodiments of the present disclosure, as shown in FIGS. **3** and **4**, multiple battery cells **11** are arranged along the first direction X; and the battery pack **10** may further comprise a second thermal management component **12**, wherein one second thermal management component **12** is arranged between every two adjacent battery cells **11** along the first direction X, and the second thermal management component **12** is configured to manage a temperature of the battery cells **11**.

[0131] The multiple battery cells **11** of the battery pack **10** are arranged inside the box **40**. The battery pack **10** includes multiple rows of battery cells **11** arranged along the first direction X, and

each row of battery cells **11** includes multiple battery cells **11** arranged along the second direction Y. Of course, in other embodiments, the battery pack **10** may also only include multiple battery cells **11** arranged along the first direction X, that is, the battery pack **10** only includes one column of multiple battery cells **11** arranged along the first direction X.

[0132] Along the first direction X, one second thermal management component **12** is installed between every two adjacent battery cells **11**, and plays a role in heat exchange with the battery cell **11** to manage the temperature of the battery cell **11**. For example, the second thermal management component **12** can be a harmonica-shaped tube or the like, and temperature regulation for the battery cell **11** can be achieved through the heat exchange medium contained within the second thermal management component **12**. The specific structure of the second thermal management component **12** can be found in relevant technologies and will not be repeated here.

[0133] It should be noted that the second thermal management component **12** and the first thermal management component **20** can have the same structure or different structures.

[0134] Multiple battery cells **11** are arranged along the first direction X, and a second thermal management component **12** is also provided between every two adjacent battery cells **11** to manage the temperature of multiple battery cells **11** in the battery pack **10** through the cooperation of the first thermal management component **20** and the second thermal management components **12**, which is conducive to alleviating the temperature rise of the battery **100** during use. In addition, due to the thermal insulating member **30** provided on one side of the first thermal management component **20** facing away from multiple battery cells **11** of the battery pack **10** in the first direction X, the phenomenon of heat exchange between the first thermal management component **20** and other components can be reduced. This ensures that the temperature management effects of the first thermal management component **20** and second thermal management component **12** on the battery cells **11** are the same, thereby reducing the temperature difference between multiple battery cells **11** in the battery pack **10**. This is beneficial for ensuring temperature to be even in various areas inside the battery **100**, thereby effectively improving the use performance and service life of the battery **100**.

[0135] In some embodiments, a thickness of the second thermal management component **12** in the first direction X is greater than a thickness of the first thermal management component **20** in the first direction X.

[0136] The thickness of the second thermal management component **12** in the first direction X is greater than the thickness of the first thermal management component **20** in the first direction X, that is, the size of the second thermal management component **12** in the first direction X is greater than the size of the first thermal management component **20** in the first direction X, so that the space inside the second thermal management component **12** for accommodating the heat exchange medium can be larger than the space inside the first thermal management component **20** for accommodating the heat exchange medium, thereby achieving a heat transfer capacity of the second thermal management component **12** greater than that of the first thermal management component **20**.

[0137] By setting the thickness of the second thermal management component **12** to be greater than that of the first thermal management component **20**, the heat transfer capacity of the second thermal management component **12** is greater than that of the first thermal management component **20**. Since the second thermal management component **12** located between two adjacent battery cells **11** in the first direction X needs to exchange heat with the battery cells **11** on both sides, while the first thermal management component **20** only needs to exchange heat with the battery cell **11** on one side, batteries **100** with this structure can meet the heat transfer needs of battery cells **11** in different regions, thereby improving the temperature uniformity in various areas inside the battery **100**.

[0138] According to some embodiments of the present disclosure, as shown in FIG. 3, the battery **100** comprises multiple battery packs **10**, and each of the battery packs **10** is equipped with the first thermal management component **20** and the thermal insulating member **30**.

[0139] Each battery pack **10** is equipped with a first thermal management component **20** and a thermal insulating member **30**, that is, each battery pack **10** is equipped with a first thermal management component **20**, and the first thermal management component **20** is set on one side of the corresponding battery pack **10** in the first direction X. At the same time, each thermal insulating member **30** is set on the side of the corresponding first thermal management component **20** that faces away from the corresponding battery pack **10** in the first direction X.

[0140] The battery **100** is provided with multiple battery packs **10**, and each of the battery packs **10** is equipped with the first thermal management component **20** and the thermal insulating member **30**, which can effectively ensure the temperature management function inside the battery **100** while improving the energy storage capacity of the battery **100**, thereby facilitating the improvement of the use performance and service life of the battery **100**.

[0141] In some embodiments, referring to FIG. 3 continuously, the multiple battery packs **10** are arranged at intervals along the first direction X. Of course, in other embodiments, the multiple battery packs **10** can also be arranged at intervals along the second direction Y.

[0142] By arranging multiple battery packs **10** at intervals along the first direction X to facilitate assembly of multiple battery packs **10**, it is beneficial to reduce the assembly difficulty of battery **100**.

[0143] According to some embodiments of the present disclosure, the embodiments of the present disclosure also provide an electrical equipment, including a battery **100** in any one of the above solutions, and the battery **100** is used to provide electrical energy to the electrical equipment.

[0144] The electrical equipment can be any of the aforementioned equipment or systems that use battery **100**.

[0145] According to some embodiments of the present disclosure, as shown in FIGS. 2 to 7, the present disclosure provides a battery **100** comprising a battery pack **10**, a first thermal management component **20**, and a thermal insulating member **30**. The battery pack **10** includes a second thermal management component **12** and multiple battery cells **11** arranged along the first direction X, and one second thermal management component **12** is provided between every adjacent two battery cells **11**. The second thermal management component **12** is configured to manage the temperature of the battery cells **11**. Along the first direction X, both sides of the battery pack **10** are equipped with first thermal management components **20**, which have a first surface **21** facing away from the battery pack **10**. The first thermal management components **20** are configured to manage the temperature of the battery pack **10**. The thermal insulating member **30** is located on the side of the first thermal management component **20** facing away from the battery pack **10** in the first direction X. The thermal insulating member **30** is connected to the first surface **21** and covers at least a part of the first surface **21**. The thermal insulating member **30** is a thermal insulating pad, and the material of the thermal insulating member **30** has a thermal conductivity coefficient of k , satisfying a condition of $k \leq 0.5 \text{ W/(mK)}$. A thickness of the thermal insulating member **30** along the first direction X is $D_{\text{sub.1}}$, satisfying a condition of $0.1 \text{ mm} \leq D_{\text{sub.1}} \leq 5 \text{ mm}$. There are multiple battery packs **10**, which are arranged at intervals along the first direction X. Each battery pack **10** is equipped with first thermal management component **20** on both sides in the first direction X, and each first thermal management component **20** is equipped with thermal insulating member **30** on the side facing away from the corresponding battery pack **10** in the first direction X.

[0146] It should be noted that in the absence of conflicts, the embodiments and the features in the embodiments in the present disclosure can be combined with each other.

[0147] The above is only preferred embodiments of the present disclosure and is not intended to limit it. For those skilled in the art, the present disclosure may undergo various modifications and variations. Any modifications, equivalent replacements, improvements, etc. made within the spirit and principles of the present disclosure shall be included within the scope of protection of the present disclosure.

Claims

1. A battery, comprising: a battery pack, comprising a plurality of battery cells; a first thermal management component, wherein along a first direction, the first thermal management component is arranged on one side of the battery pack, and the first thermal management component is configured to manage a temperature of the battery pack; and a thermal insulating member, wherein along the first direction, the thermal insulating member is positioned on one side of the first thermal management component facing away from the battery pack.
2. The battery according to claim 1, wherein along the first direction, two sides of the battery pack comprise the first thermal management component, and each of the first thermal management component comprises the thermal insulating member on a side facing away from the battery pack.
3. The battery according to claim 1, wherein the first thermal management component comprises a first surface facing away from the battery pack along the first direction; and the thermal insulating member is connected to the first surface and covers at least a part of the first surface.
4. The battery according to claim 1, wherein the thermal insulating member is a thermal insulating pad, and a material of the thermal insulating member comprises a thermal conductivity coefficient of k , which satisfies $k \leq 0.5 \text{ W/(mK)}$.
5. The battery according to claim 4, wherein the thermal insulating member comprises a second surface facing away from the first thermal management component along the first direction, and a plurality of grooves are located on the second surface.
6. The battery according to claim 5, wherein along a second direction, the grooves penetrate through both ends of the thermal insulating member, and the second direction is perpendicular to the first direction.
7. The battery according to claim 4, wherein along the first direction, a thickness of the thermal insulating member is $D_{\text{sub.1}}$, satisfying a condition of $0.1 \text{ mm} \leq D_{\text{sub.1}} \leq 5 \text{ mm}$.
8. The battery according to claim 1, wherein a thermal insulating chamber is formed inside the thermal insulating member.
9. The battery according to claim 8, wherein along the first direction, a wall thickness of the thermal insulating chamber is $D_{\text{sub.2}}$, satisfying a condition of $0.1 \text{ mm} \leq D_{\text{sub.2}} \leq 3 \text{ mm}$.
10. The battery according to claim 8, wherein the thermal insulating member comprises a reinforcing part, wherein the reinforcing part is located within the thermal insulating chamber; and along the first direction, the reinforcing part is connected to two chamber wall surfaces of the thermal insulating chamber that are facing each other.
11. The battery according to claim 8, wherein the thermal insulating member and the first thermal management component are integrally formed.
12. The battery according to claim 1, wherein along the first direction, an accommodating groove is located on one side of the first thermal management component facing away from the battery pack, and the accommodating groove is configured to hold at least a part of the thermal insulating member.
13. The battery according to claim 1, wherein the plurality of battery cells are arranged along the first direction; and the battery pack further comprises a second thermal management component, wherein one second thermal management component is arranged between every two adjacent battery cells along the first direction, and the second thermal management component is configured to manage a temperature of the plurality of battery cells.
14. The battery according to claim 13, wherein a thickness of the second thermal management component in the first direction is greater than a thickness of the first thermal management component in the first direction.
15. The battery according to claim 1, wherein the battery comprises: a plurality of battery packs, wherein each of the plurality of battery packs is equipped with the first thermal management

component and the thermal insulating member.

16. The battery according to claim 15, wherein the plurality of battery packs are arranged at intervals along the first direction.

17. An electrical equipment, comprising a battery that is configured to provide an electric energy, the battery comprising: a battery pack, comprising a plurality of battery cells; a first thermal management component, wherein along a first direction, the first thermal management component is arranged on one side of the battery pack, and the first thermal management component is configured to manage a temperature of the battery pack; and a thermal insulating member, wherein along the first direction, the thermal insulating member is positioned on one side of the first thermal management component facing away from the battery pack.

18. The electrical equipment according to claim 17, wherein along the first direction, two sides of the battery pack comprise the first thermal management component, and each of the first thermal management component comprises the thermal insulating member on a side facing away from the battery pack.

19. The electrical equipment according to claim 17, wherein the first thermal management component comprises a first surface facing away from the battery pack along the first direction; and the thermal insulating member is connected to the first surface and covers at least a part of the first surface.

20. The electrical equipment according to claim 17, wherein the thermal insulating member is a thermal insulating pad, and a material of the thermal insulating member comprises a thermal conductivity coefficient of k , which satisfies $k \leq 0.5 \text{ W/(mK)}$.
